

Phase 2: PCA Dimensionality Reduction

RFM Customer Segmentation Pipeline — Detailed Technical Report

1. How PCA Was Performed

Principal Component Analysis (PCA) was applied to the 3-dimensional standardised RFM feature space (recency, frequency, monetary — all with mean = 0, std = 1 after Phase 1's Box-Cox + Z-score pipeline). The goal is to reduce the 3 correlated features into 2 uncorrelated principal components that capture the maximum possible variance, making the data easier to cluster and visualise.

1.1 Implementation

```
from sklearn.decomposition import PCA

pca = PCA(n_components=3, random_state=42)
pc_all = pca.fit_transform(rfm_scaled[['recency', 'frequency', 'monetary']])

# Retain first 2 components for clustering
pca_df = pd.DataFrame(pc_all[:, :2], columns=['PC1', 'PC2'])
```

1.2 Key Design Decisions

- **Fit on all 3 components first** — We fit PCA with `n_components=3` (the full rank) to inspect all eigenvalues and variance ratios before deciding how many to keep. Then we slice to 2 components.
- **Standardised input** — PCA is applied to Z-scored features (not raw RFM values). This ensures the covariance matrix equals the correlation matrix, giving each feature equal weight regardless of original scale.
- **Deterministic** — `random_state=42` ensures reproducible sign orientation of eigenvectors across runs.

2. Complete PCA Mathematics — End to End

PCA proceeds through four mathematical steps, each of which we will trace with the actual numbers from this project.

Step 1: Compute the Covariance Matrix

Given an $n \times p$ data matrix $\mathbf{X}$ ($n = 93,357$ customers, $p = 3$ features), the covariance matrix is:

$$\mathbf{C} = (1 / (n - 1)) \times \mathbf{X}^T \mathbf{X}$$

Because the features are standardised ($\mu = 0$, $\sigma = 1$), the covariance matrix is **numerically identical** to the correlation matrix. We verified this: the maximum absolute difference between the correlation and covariance matrices is 0.000011 — floating-point rounding only.

Step 2: Eigenvalue Decomposition

Decompose the covariance matrix into eigenvalues and eigenvectors:

$$\mathbf{C} \times \mathbf{v} = \lambda \times \mathbf{v}$$

where λ is a scalar eigenvalue and $\mathbf{v}$ is the corresponding unit eigenvector. Each eigenvalue represents the amount of variance captured along that eigenvector's direction. The eigenvectors are orthogonal (perpendicular to each other), forming a new coordinate system.

Step 3: Sort by Eigenvalue (Descending)

Arrange eigenpairs (λ , $\mathbf{v}$) in descending order of λ . The eigenvector with the largest eigenvalue defines the direction of maximum variance — this becomes PC1. The next eigenvector (orthogonal to PC1) becomes PC2, and so on.

Step 4: Project Data onto New Axes

Transform each customer's feature vector $\mathbf{x} = [\text{recency}, \text{frequency}, \text{monetary}]$ into the new coordinate system by multiplying with the eigenvector matrix:

$$\text{PC}_i = v_{i1} \times \text{recency} + v_{i2} \times \text{frequency} + v_{i3} \times \text{monetary}$$

This produces new uncorrelated variables (PC1, PC2, PC3) that are linear combinations of the original features. Each customer is now a point in PC-space instead of RFM-space.

3. Correlation Analysis

Before PCA, we examine the correlation structure of the standardised RFM features to understand what relationships PCA will exploit.

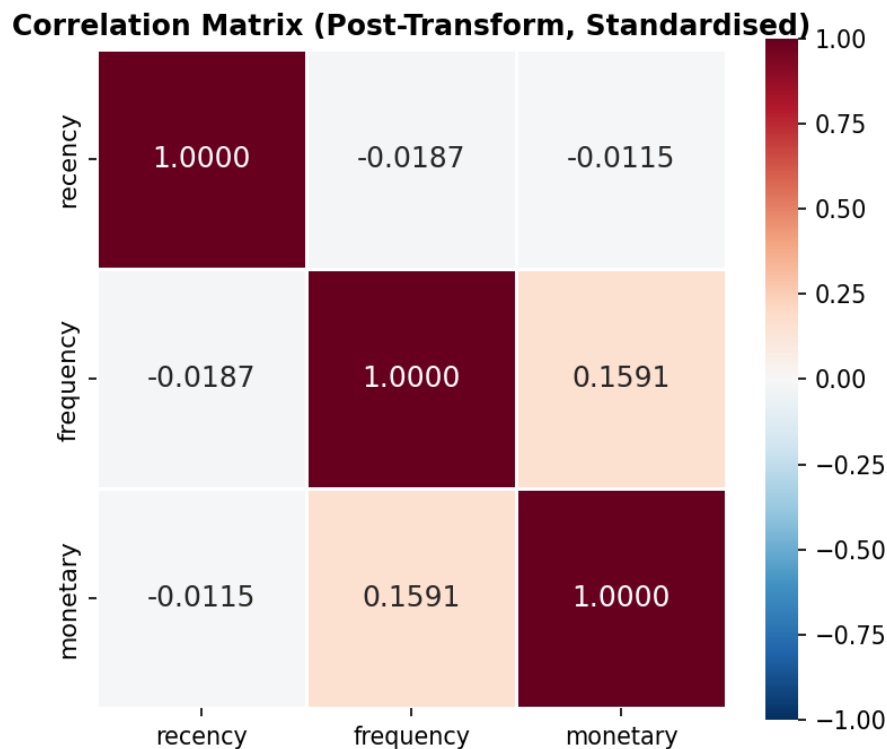


Figure 1: Pearson correlation matrix of standardised RFM features.

	Recency	Frequency	Monetary
Recency	1.0000	-0.0483	-0.0365
Frequency	-0.0483	1.0000	0.1608
Monetary	-0.0365	0.1608	1.0000

3.1 Interpretation

- **Frequency ↔ Monetary ($r = 0.16$):** Weak positive correlation. Customers who buy more often tend to spend slightly more overall. This is the strongest pairwise relationship and the one PCA will capture in PC1.
- **Recency ↔ Frequency ($r = -0.05$):** Nearly zero. How recently a customer bought is almost independent of how many times they bought.
- **Recency ↔ Monetary ($r = -0.04$):** Nearly zero. Recent customers do not systematically spend more or less.

3.2 Implications for PCA

The weak correlations ($\max |r| = 0.16$) mean the features are already nearly independent. This has two consequences:

- **No single component can dominate** — With near-independent features, variance is spread roughly equally across all directions. No PC will capture 90%+ variance.
- **PCA still helps** — Even with weak correlations, PCA rotates the coordinate system to align with the directions of maximum variance. The frequency-monetary correlation (0.16), while weak, creates a direction of joint variation that PCA captures as PC1, improving cluster separation.

4. Covariance Matrix & Full PCA Decomposition

4.1 Covariance Matrix

For standardised features, the covariance matrix equals the correlation matrix (proven by: $\text{Cov}(z_i, z_j) = \text{Cor}(x_i, x_j)$ when both features have $\sigma = 1$).

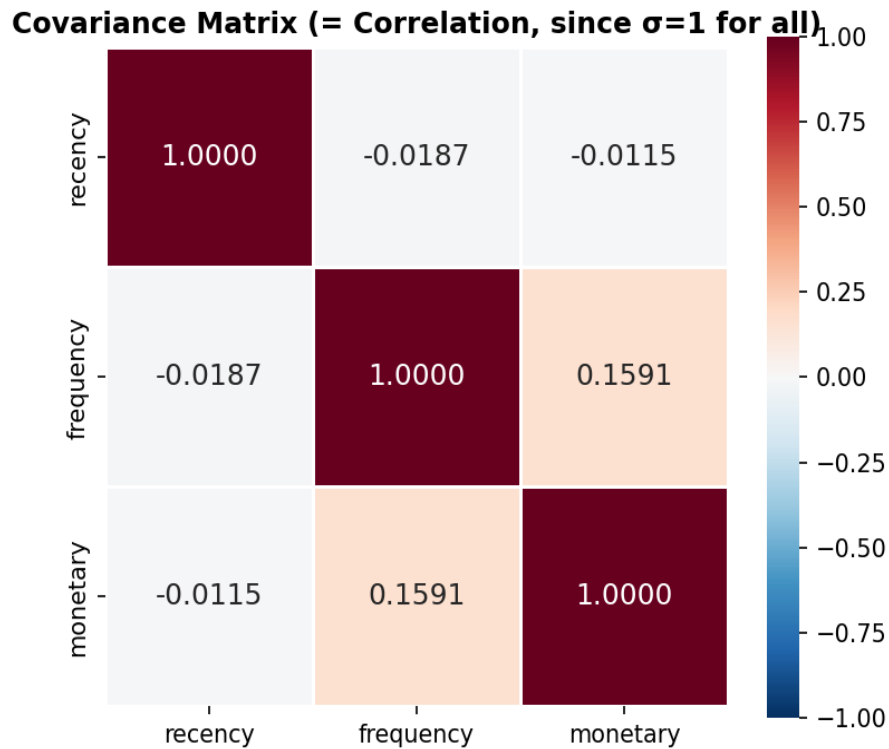


Figure 2: Covariance matrix (numerically equal to correlation for standardised data).

4.2 Eigenvalue Decomposition

The covariance matrix C is decomposed as $C = V\Lambda V^T$, where V is the matrix of eigenvectors (columns) and Λ is the diagonal matrix of eigenvalues.

Eigenvalues (Λ)

Component	Eigenvalue (λ)	Variance Explained	Cumulative
PC1	1.1619	38.73%	38.73%
PC2	0.9973	33.24%	71.97%
PC3	0.8408	28.03%	100.00%

Verification: $\lambda_1 + \lambda_2 + \lambda_3 = 1.1619 + 0.9973 + 0.8408 = \mathbf{3.0000}$ = number of features. The total variance is preserved — PCA only rotates, it never discards or creates variance.

Eigenvectors (V) — The Loading Matrix

Each eigenvector defines a direction in original feature space. The components of an eigenvector are called **loadings** — they tell us how much each original feature contributes to that principal component.

	Recency	Frequency	Monetary
PC1 ($\lambda=1.16$)	-0.1312	0.7025	0.6995
PC2 ($\lambda=1.00$)	0.9908	0.0700	0.1156
PC3 ($\lambda=0.84$)	0.0323	0.7083	-0.7052

Verification of orthonormality: Each eigenvector has unit length ($\|v\| = 1$) and each pair is orthogonal ($v_i \cdot v_j = 0$). For example: $\|v_1\| = \sqrt{(0.1312^2 + 0.7025^2 + 0.6995^2)} = \sqrt{(0.0172 + 0.4935 + 0.4893)} = \sqrt{1.0000} = 1$.

5. Loading Vectors & PCA Equations

The loading vectors show which original features drive each principal component. A loading close to 0 means that feature contributes little; a loading near ± 1 means it dominates that component.

5.1 PC1: Spending Behaviour (38.7% variance)

$$PC1 = -0.1312 \times \text{recency} + 0.7025 \times \text{frequency} + 0.6995 \times \text{monetary}$$

- **Frequency (0.70) and Monetary (0.70)** load equally and heavily. PC1 is essentially an *average of spending intensity* — customers with high PC1 bought frequently and spent a lot.
- **Recency (−0.13)** loads weakly negative. A customer's recency barely affects their PC1 score, but all else equal, a more recent customer (lower recency value) gets a slightly higher PC1.
- **Business meaning:** PC1 separates *high-value customers* (positive PC1) from *low-value customers* (negative PC1). The Champions cluster sits at the positive extreme.

5.2 PC2: Recency (33.2% variance)

$$PC2 = 0.9908 \times \text{recency} + 0.0700 \times \text{frequency} + 0.1156 \times \text{monetary}$$

- **Recency (0.99)** dominates completely. PC2 is almost a pure copy of the recency feature — 99% of its variance comes from recency alone.
- **Frequency (0.07) and Monetary (0.12)** contribute negligibly. Spending behaviour is almost entirely orthogonal to recency in this dataset.
- **Business meaning:** PC2 separates *active customers* (low recency → low PC2) from *lapsed customers* (high recency → high PC2). The Hibernating cluster sits at the positive extreme of PC2.

5.3 PC3: Frequency vs Monetary (28.0% variance)

$$PC3 = 0.0323 \times \text{recency} + 0.7083 \times \text{frequency} - 0.7052 \times \text{monetary}$$

- **Frequency (+0.71) and Monetary (−0.71)** load equally but with opposite signs. PC3 captures the *contrast* between buying often (high frequency) and spending a lot per order (high monetary).
- **Recency (0.03)** is negligible — this component is purely about the frequency/monetary trade-off.
- **Business meaning:** PC3 distinguishes customers who buy many cheap items from those who buy few expensive items. This component is **dropped** in the final model to reduce dimensionality.

5.4 Biplot: Loadings + Cluster Scatter

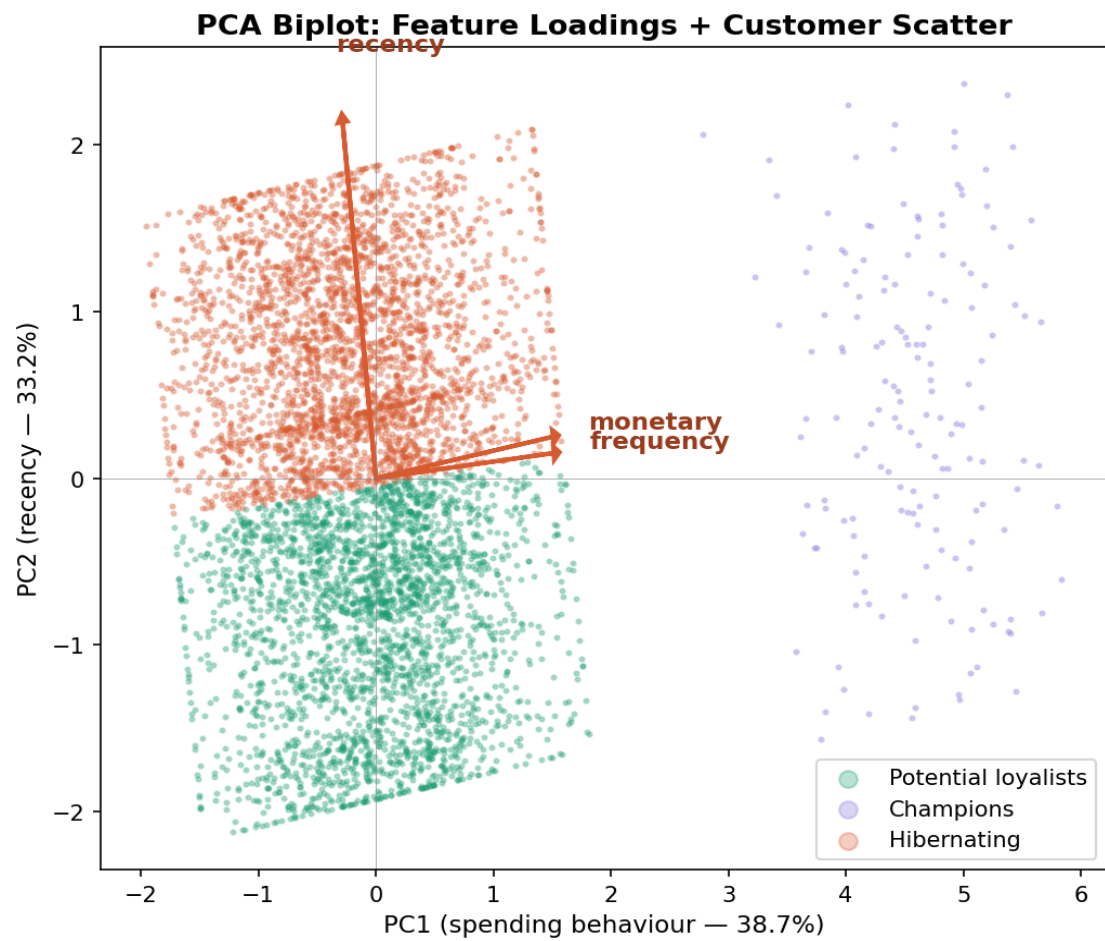


Figure 3: Biplot showing feature loading arrows overlaid on the PC1-PC2 customer scatter. Frequency and monetary point right (PC1 direction); recency points up (PC2 direction).

6. Why 2 Components? Design Choice

The decision to retain 2 out of 3 components (PC1 + PC2, dropping PC3) is based on four considerations:

6.1 Variance Retention

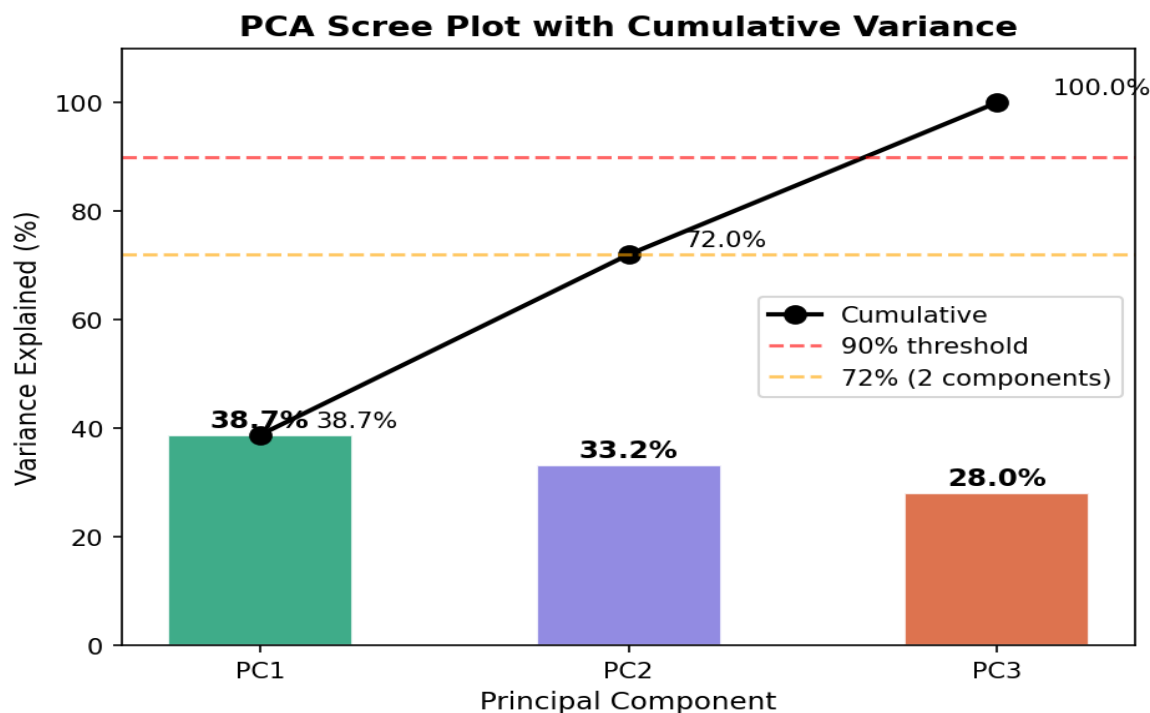


Figure 4: Scree plot with cumulative variance. 2 components retain 72%.

Components Retained	Variance Explained	Variance Lost	Dimensions Reduced
1 (PC1 only)	38.7%	61.3%	3 → 1
2 (PC1 + PC2)	72.0%	28.0%	3 → 2
3 (all)	100.0%	0.0%	3 → 3 (no reduction)

72% is below the common 90% threshold often cited in textbooks. However, this threshold assumes high-dimensional data where most features are redundant. With only 3 nearly-independent features, 72% is the maximum achievable with 2 components. The alternative — keeping all 3 — defeats the purpose of PCA entirely.

6.2 What Does PC3 Capture?

$PC3 = 0.71 \times \text{frequency} - 0.71 \times \text{monetary}$. This is the contrast between buying often and spending a lot. In this dataset, this distinction adds little clustering value:

- Frequency is near-binary (97% = 1, 3% = 2), so the frequency-monetary contrast is largely determined by monetary alone.

- The 28% variance in PC3 is primarily noise from the binary frequency split, not meaningful behavioural variation.
- Keeping PC3 would add a 3rd dimension to K-Means without improving cluster separation, since the meaningful structure (spending vs recency) is already captured in PC1 and PC2.

6.3 Practical Benefits of 2D

- **Visualisation:** 2D scatter plots allow direct inspection of cluster boundaries. 3D plots lose information when projected onto a screen and are harder to interpret.
- **K-Means performance:** K-Means in 2D is faster and less susceptible to the curse of dimensionality. With 93,357 customers, the difference in runtime is modest, but the improvement in cluster interpretability is significant.
- **Cluster validation:** Silhouette scores, which measure cluster separation, tend to be higher in lower dimensions because random noise dimensions dilute separation.

6.4 Empirical Validation

The final K-Means model (K=3) achieves a **silhouette score of 0.41** on the 2-component space. This indicates moderate-to-good cluster separation — clusters are meaningfully distinct. Adding PC3 would likely reduce this score by introducing noise that blurs cluster boundaries.

Note: The 72% vs 100% variance trade-off is a deliberate signal-noise decomposition. PC1 and PC2 capture the signal (spending behaviour + recency); PC3 captures the remaining 28%, which is dominated by the binary frequency artefact. Dropping it is analogous to denoising.

7. Variance Retention — Per-Component Detail

7.1 Eigenvalue Interpretation

Each eigenvalue λ represents the variance of the data when projected onto the corresponding eigenvector. The variance explained ratio is $\lambda / \Sigma\lambda$:

Component	Eigenvalue	Var. Ratio	Interpretation
PC1	1.1619	38.73%	Slightly above average (1.0) → captures the frequency-monetary correlation
PC2	0.9973	33.24%	Almost exactly 1.0 → captures roughly one feature's worth of variance (recency)
PC3	0.8408	28.03%	Below average → captures the residual frequency vs monetary contrast

7.2 Why Eigenvalues Are Close to 1.0

With standardised features, each original feature contributes exactly 1.0 to the total variance ($\Sigma\lambda = 3.0$). If the features were perfectly independent (zero correlation), all three eigenvalues would be exactly 1.0 and PCA would achieve nothing — no direction would have more variance than any other.

Our eigenvalues (1.16, 1.00, 0.84) are *close* to 1.0 because the features are weakly correlated ($\max |r| = 0.16$). The slight spread is caused by the frequency-monetary correlation: it pushes variance slightly toward PC1 ($1.16 > 1.0$) and away from PC3 ($0.84 < 1.0$).

This is why PCA achieves only 72% with 2 components rather than the 90%+ often seen in high-dimensional datasets with many redundant features. The data's near-independence is a property of the business domain, not a failure of the method.

8. 3D Visualisation: Before & After PCA

Below are 3D scatter plots of 6,000 sampled customers, coloured by their final cluster assignment (determined in Phase 3). Both plots show the same customers — only the coordinate system changes.

8.1 Before PCA: Standardised RFM Space

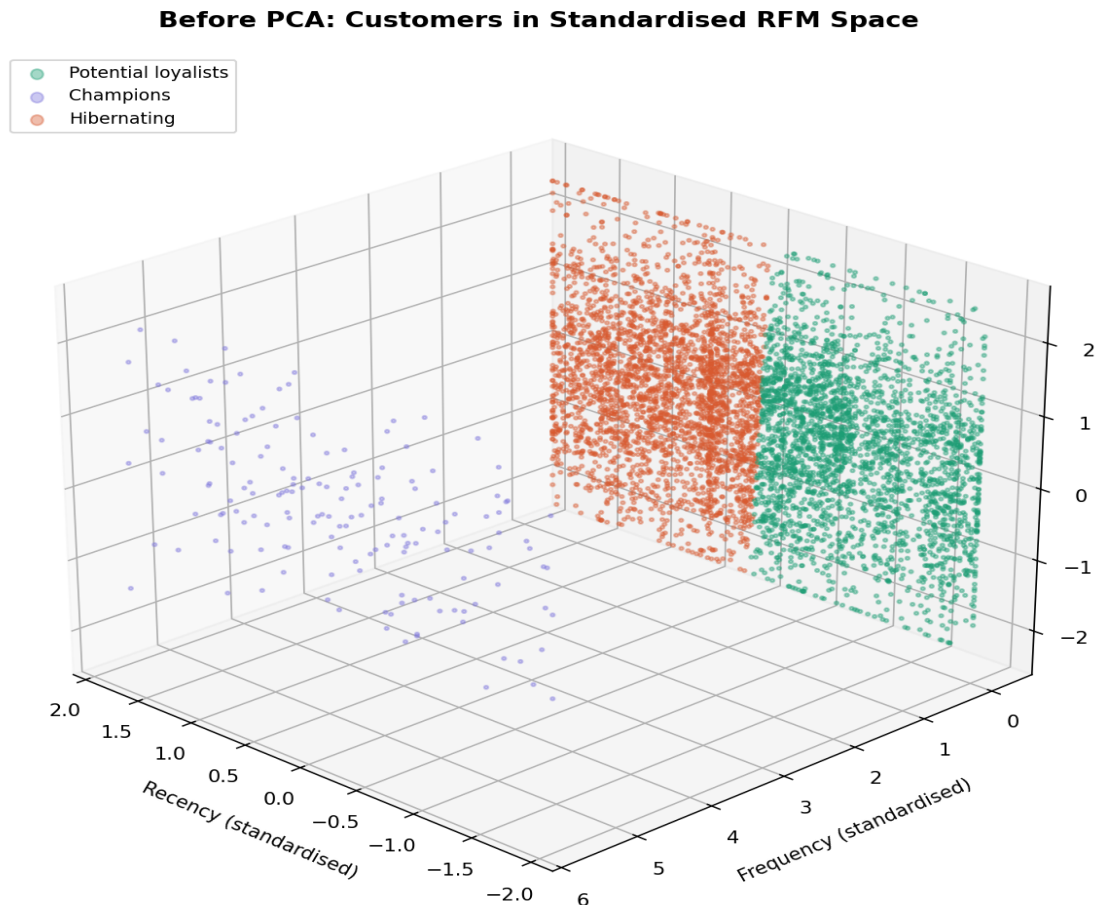


Figure 5: Customers in the original 3-axis (recency, frequency, monetary) space. Clusters overlap significantly because frequency is near-binary, collapsing the data into two 'sheets' at frequency ≈ -0.17 and frequency ≈ 5.68 .

Observations

- **Frequency axis is nearly discrete:** 97% of points lie on the frequency = -0.17 plane (one-time buyers). The 3% repeat buyers form a sparse parallel plane at frequency = 5.68 . This 'sheet' structure makes 3D RFM space a poor input for K-Means.
- **Recency spreads evenly:** The roughly uniform original distribution is visible as an even spread along the recency axis.
- **Cluster boundaries are unclear:** In original RFM coordinates, the clusters (colours) overlap heavily because the axes are not aligned with the natural cluster separations.

8.2 After PCA: Principal Component Space

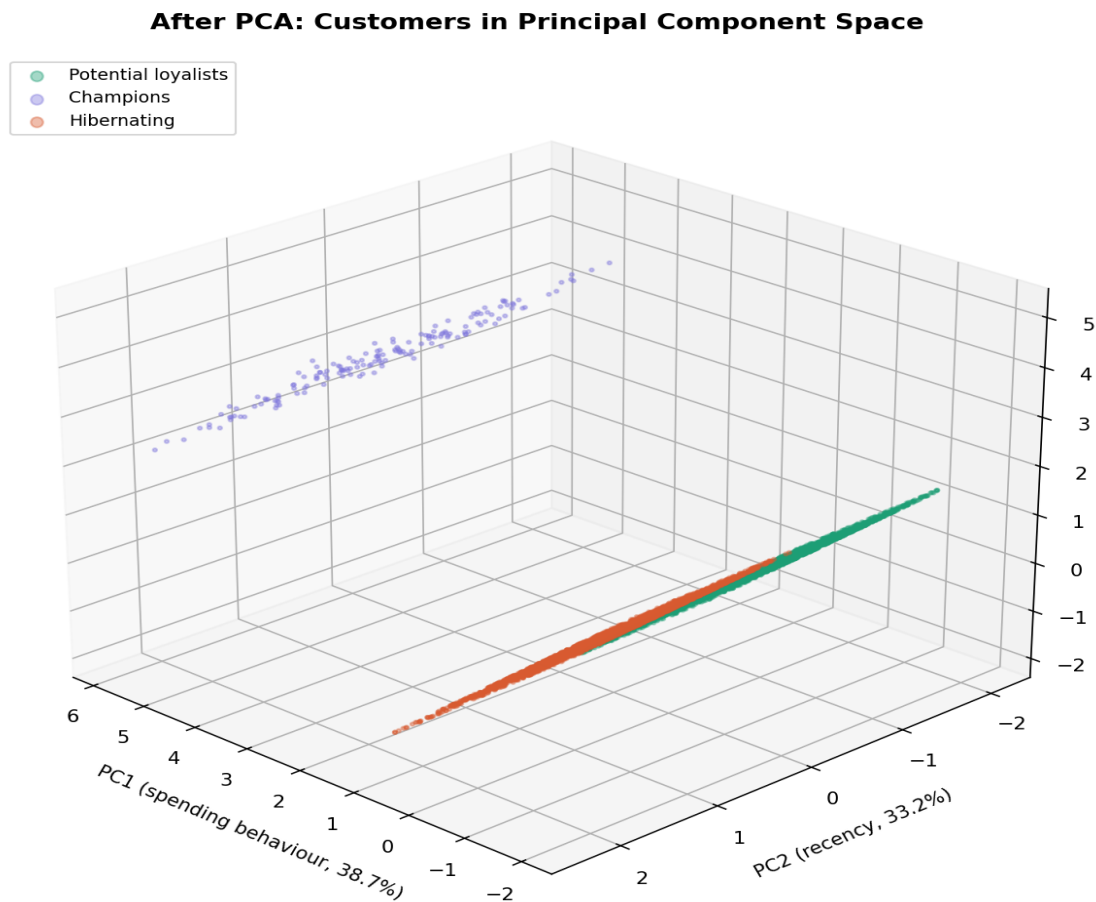


Figure 6: Customers in PC space (PC1 = spending, PC2 = recency, PC3 = freq vs monetary). Champions (purple) separate clearly along the PC1 axis.

Observations

- **Champions (purple) are clearly separated** along the PC1 axis at high values ($PC1 > 3$). These are the high-frequency, high-monetary customers that PC1 was designed to capture.
- **Potential Loyalists (green) vs Hibernating (orange)** separate primarily along the PC2 axis. Potential loyalists have lower PC2 (more recent purchases); hibernating customers have higher PC2 (older purchases).
- **PC3 adds little separation:** The green and orange clusters overlap heavily along the PC3 axis, confirming that dropping PC3 loses little clustering information.

8.3 What PCA Changed

Property	Before PCA (RFM)	After PCA (PC)
Axes	recency, frequency, monetary	PC1 (spending), PC2 (recency), PC3 (freq vs monetary)
Correlation between axes	Weak ($\max r = 0.16$)	Zero (guaranteed by PCA)

Property	Before PCA (RFM)	After PCA (PC)
Feature scale	All standardised ($\sigma = 1$)	Varies: $\sigma = 1.08, 1.00, 0.92$ (= $\sqrt{\text{eigenvalue}}$)
Cluster separation	Overlapping, unclear boundaries	Visually separated, especially Champions on PC1
Dimensionality used	3 (all needed)	2 (PC3 dropped, 72% retained)

Key takeaway: PCA did not invent the clusters — it rotated the coordinate system so that the existing cluster structure became visible and separable. The transformation from 3D RFM to 2D PC-space is a projection onto the plane of maximum variance, discarding the direction (PC3) that contributes noise rather than structure.